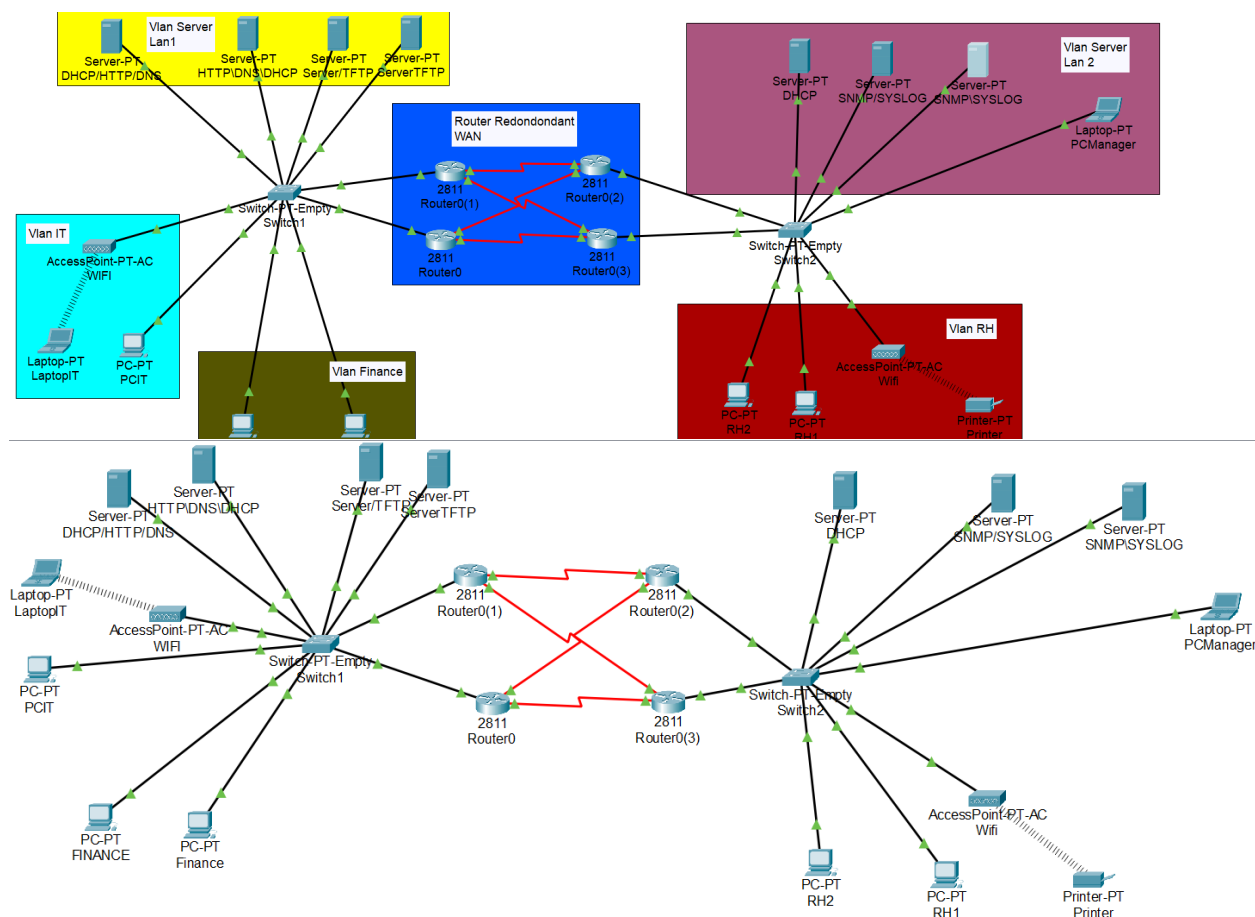


Guide de configuration réseau - WAN Simulé

I/ Objectif du projet

La PME « NeoEdu » a deux agences distantes reliées par une liaison WAN simulée. Elle souhaite un réseau hautement disponible avec redondance, VLANs par département, automatisation des sauvegardes et supervision avancée (SNMP, syslog). Le rôle est d'assurer sécurité, performance et diagnostic automatisé, et de produire une documentation complète.

Schéma du réseau



II/ Configuration des équipements Cisco

Attribution d'une adresse ip aux servers

```
IP Server DHCP = 192.168.80.4 255.255.255.0
IP Server SNMP/SYSLOG = 192.168.80.5 255.255.255.0
IP Server SNMP\SYSLOG = 192.168.80.6 255.255.255.0
IP Server DHCP/HTTP/DNS = 192.168.90.4 255.255.255.0
IP Server DHCP\HTTP\DNS = 192.168.90.5 255.255.255.0
IP ServerTFTP = 192.168.90.7 255.255.255.0
IP Server/TFTP = 192.168.90.6 255.255.255.0
```

Configuration des switchs

#Switch 1

```
Vlan 3
name Finance
Vlan 4
name IT
Vlan 90
name SERVER
```

```
interface g0/1
switchport mode trunk
switchport trunk Vlan 3,4,90
no shutdown
exit
```

```
interface g2/1
switchport mode access
switchport access Vlan 4
no shutdown
exit
```

```
interface g3/1
switchport mode access
switchport access Vlan 90
no shutdown
exit
```

```
interface g4/1
switchport mode access
switchport access Vlan 90
no shutdown
exit
```

```
interface g5/1
switchport mode access
switchport access Vlan 90
no shutdown
exit
```

```
interface g6/1
switchport mode access
switchport access Vlan 90
no shutdown
exit
```

```
interface g7/1
switchport mode access
switchport access Vlan 4
no shutdown
exit
```

```
interface g8/1
switchport mode access
switchport access Vlan 3
no shutdown
exit
```

```
interface g9/1
switchport mode access
switchport access Vlan 3
```

```
no shutdown
exit
write memory

#Switch 2

Vlan 2
name RH
Vlan 80
name Server

interface g0/1
switchport mode trunk
switchport trunk Vlan 2,80
no shutdown
exit

interface g3/1
switchport mode access
switchport access Vlan 2
no shutdown
exit

interface g4/1
switchport mode access
switchport access Vlan 2
no shutdown
exit

interface g9/1
switchport mode access
switchport access Vlan 2
no shutdown
exit

interface g5/1
switchport mode access
switchport access Vlan 80
no shutdown
exit

interface g6/1
switchport mode access
switchport access Vlan 80
no shutdown
exit

interface g7/1
switchport mode access
switchport access Vlan 80
no shutdown
exit

interface g8/1
switchport mode access
switchport access Vlan 80
no shutdown
exit
write memory
```

Device Name: Switch1
Device Model: Switch-PT-Empty
Hostname: Switch

Port	Link	VLAN	IP Address	MAC Address
GigabitEthernet0/1	Up	--	--	0001.9633.AEC0
GigabitEthernet1/1	Up	--	--	0090.2B7B.51DE
GigabitEthernet2/1	Up	4	--	00E0.A3EA.45D5
GigabitEthernet3/1	Up	90	--	00D0.BA09.D8A9
GigabitEthernet4/1	Up	90	--	0001.C9D2.D4B0
GigabitEthernet5/1	Up	90	--	0002.170D.E8E5
GigabitEthernet6/1	Up	90	--	0090.21BB.6E06
GigabitEthernet7/1	Up	4	--	00D0.BC8D.E49B
GigabitEthernet8/1	Up	3	--	0050.0FBB.6861
GigabitEthernet9/1	Up	3	--	00E0.F9BC.3952
Vlan1	Down	1	<not set>	0007.EC96.6524

Physical Location: Intercity > Home City > Corporate Office > Main Wiring Closet > Rack > Switch1

Device Name: Switch2
Device Model: Switch-PT-Empty
Hostname: Switch

Port	Link	VLAN	IP Address	MAC Address
GigabitEthernet0/1	Up	--	--	0090.0CC6.E62B
GigabitEthernet1/1	Down	1	--	0007.EC96.645A
GigabitEthernet2/1	Up	--	--	00D0.BC98.99A8
GigabitEthernet3/1	Up	2	--	00E0.B0CC.84E3
GigabitEthernet4/1	Up	2	--	00D0.97E2.B6C2
GigabitEthernet5/1	Up	80	--	00D0.D378.C684
GigabitEthernet6/1	Up	80	--	0000.0C37.98B9
GigabitEthernet7/1	Up	80	--	0006.2A65.A939
GigabitEthernet8/1	Up	80	--	000A.F32A.5D5C
GigabitEthernet9/1	Up	2	--	0090.0C5E.28BE
Vlan1	Down	1	<not set>	0005.5E9D.5792

Physical Location: Intercity > Home City > Corporate Office > Main Wiring Closet > Rack > Switch2

Configuration des Routers

Router 1

```
enable
configure terminal
interface fa0/0
ip address 192.168.1.2 255.255.255.0
no shutdown
exit
```

#Router on stick

#Vlan Finance

```
interface fa0/0.3
encapsulation dot1q 3
ip address 192.168.3.2 255.255.255.0
ip helper-address 192.168.90.5
ip helper-address 192.168.90.4
no shutdown
exit
```

#Vlan IT

```
interface fa0/0.4
encapsulation dot1q 4
ip address 192.168.4.2 255.255.255.0
ip helper-address 192.168.90.5
ip helper-address 192.168.90.4
no shutdown
exit
```

#Vlan SERVER

```
interface fa0/0.90
encapsulaiton dot1q 90
ip address 192.168.90.2 255.255.255.0
```

```
no shutdown
```

```
exit
```

```
#Redondance Router 1 Actif
```

```
interface fa0/0
```

```
standby 1 ip 192.168.1.1
```

```
standby 1 priority 110
```

```
standby 1 preempt
```

```
no shutdown
```

```
exit
```

```
interface fa0/0.3
```

```
standby 1 ip 192.168.3.1
```

```
standby 1 priority 110
```

```
standby 1 preempt
```

```
no shutdown
```

```
exit
```

```
interface fa0/0.4
```

```
standby 1 ip 192.168.4.1
```

```
standby 1 priority 110
```

```
standby 1 preempt
```

```
no shutdown
```

```
exit
```

```
interface fa0/0.90
```

```
standby 1 ip 192.168.90.1
```

```
standby 1 priority 110
```

```
standby 1 preempt
```

```
no shutdown
```

```
exit
```

```
# Router 2
```

```
enable
```

```
configure terminal
```

```
interface fa0/0
```

```
ip address 192.168.1.3 255.255.255.0
```

```
no shutdown
```

```
exit
```

```
#Router on stick
```

```
#Vlan Finance
```

```
interface fa0/0.3
```

```
encapsulation dot1q 3
```

```
ip address 192.168.3.3 255.255.255.0
```

```
ip helper-address 192.168.90.5
```

```
ip helper-address 192.168.90.4
```

```
no shutdown
```

```
exit
```

```
#Vlan IT
```

```
interface fa0/0.4
```

```
encapsulation dot1q 4
```

```
ip address 192.168.4.3 255.255.255.0
```

```
ip helper-address 192.168.90.5
```

```
ip helper-address 192.168.90.4
```

```
no shutdown
```

```
exit
```

```
#Vlan SERVER
```

```
interface fa0/0.90
encapsulation dot1q 90
ip address 192.168.90.3 255.255.255.0
no shutdown
exit
```

#Redondance Router 2 Standby

```
interface fa0/0
standby 1 ip 192.168.1.1
standby 1 priority 100
standby 1 preempt
no shutdown
exit
```

```
interface fa0/0.3
standby 1 ip 192.168.3.1
standby 1 priority 100
standby 1 preempt
no shutdown
exit
```

```
interface fa0/0.4
standby 1 ip 192.168.4.1
standby 1 priority 100
standby 1 preempt
no shutdown
exit
```

```
interface fa0/0.90
standby 1 ip 192.168.90.1
standby 1 priority 100
standby 1 preempt
no shutdown
exit
```

Router 1 bis

```
enable
configure terminal
interface fa0/0
ip address 192.168.10.2 255.255.255.0
no shutdown
exit
```

#Router on stick

#Vlan RH

```
interface fa0/0.2
encapsulation dot1q 2
ip address 192.168.2.2 255.255.255.0
ip helper-address 192.168.90.5
ip helper-address 192.168.80.4
no shutdown
exit
```

#Vlan Server

```
interface fa0/0.80
encapsulation dot1q 80
ip address 192.168.80.2 255.255.255.0
no shutdown
```

```
exit
```

```
#Redondance Router 1bis Actif
```

```
interface fa0/0
standby 1 ip 192.168.10.1
standby 1 priority 110
standby 1 preempt
no shutdown
exit
```

```
interface fa0/0.2
standby 1 ip 192.168.2.1
standby 1 priority 110
standby 1 preempt
no shutdown
exit
```

```
interface fa0/0.80
standby 1 ip 192.168.80.1
standby 1 priority 110
standby 1 preempt
no shutdown
exit
```

```
# Router 2 bis
```

```
enable
configure terminal
interface fa0/0
ip address 192.168.10.3 255.255.255.0
no shutdown
exit
```

```
#Router on stick
```

```
#Vlan RH
```

```
interface fa0/0.2
encapsulation dot1q 2
ip address 192.168.2.3 255.255.255.0
ip helper-address 192.168.90.5
ip helper-address 192.168.80.4
no shutdown
exit
```

```
#Vlan Server
```

```
interface fa0/0.80
encapsulaiton dot1q 80
ip address 192.168.80.3 255.255.255.0
no shutdown
exit
```

```
#Redondance Router 2bis Standby
```

```
interface fa0/0
standby 1 ip 192.168.10.1
standby 1 priority 100
standby 1 preempt
no shutdown
exit
```

```

interface fa0/0.2
standby 1 ip 192.168.2.1
standby 1 priority 100
standby 1 preempt
no shutdown
exit

```

```

interface fa0/0.80
standby 1 ip 192.168.80.1
standby 1 priority 100
standby 1 preempt
no shutdown
exit

```

Device Name: Router0(1)						
Custom Device Model: 2811 IOS15						
Hostname: Router						
Port	Link	VLAN	IP Address	IPv6 Address	MAC Address	
FastEthernet0/0	Up	--	192.168.1.2/24	<not set>	0060.2F9C.3698	
FastEthernet0/0.3	Up	--	192.168.3.2/24	<not set>	0060.2F9C.3698	
FastEthernet0/0.4	Up	--	192.168.4.2/24	<not set>	0060.2F9C.3698	
FastEthernet0/0.90	Up	--	192.168.90.2/24	<not set>	0060.2F9C.3698	
FastEthernet0/1	Down	--	<not set>	<not set>	00D0.979A.186D	
Serial0/2/0	Up	--	10.0.0.1/30	<not set>	<not set>	
Serial0/2/1	Up	--	10.0.0.5/30	<not set>	<not set>	
Serial0/3/0	Down	--	<not set>	<not set>	<not set>	
Serial0/3/1	Down	--	<not set>	<not set>	<not set>	
Vlan1	Down	1	<not set>	<not set>	0090.2BB5.347E	
Physical Location: Intercity > Home City > Corporate Office > Main Wiring Closet > Rack > Router0(1)						

Device Name: Router0(2)						
Custom Device Model: 2811 IOS15						
Hostname: Router						
Port	Link	VLAN	IP Address	IPv6 Address	MAC Address	
FastEthernet0/0	Up	--	192.168.10.2/24	<not set>	0050.0F7B.36D0	
FastEthernet0/0.2	Up	--	192.168.2.2/24	<not set>	0050.0F7B.36D0	
FastEthernet0/0.80	Up	--	192.168.80.2/24	<not set>	0050.0F7B.36D0	
FastEthernet0/1	Down	--	<not set>	<not set>	0009.7C81.669E	
Serial0/2/0	Up	--	10.0.0.2/30	<not set>	<not set>	
Serial0/2/1	Up	--	10.0.0.10/30	<not set>	<not set>	
Serial0/3/0	Down	--	<not set>	<not set>	<not set>	
Serial0/3/1	Down	--	<not set>	<not set>	<not set>	
Vlan1	Down	1	<not set>	<not set>	00E0.F91C.0372	
Physical Location: Intercity > Home City > Corporate Office > Main Wiring Closet > Rack > Router0(2)						

Device Name: Router0						
Custom Device Model: 2811 IOS15						
Hostname: Router						
Port	Link	VLAN	IP Address	IPv6 Address	MAC Address	
FastEthernet0/0	Up	--	192.168.1.3/24	<not set>	0090.21DB.1301	
FastEthernet0/0.3	Up	--	192.168.3.3/24	<not set>	0090.21DB.1301	
FastEthernet0/0.4	Up	--	192.168.4.3/24	<not set>	0090.21DB.1301	
FastEthernet0/0.90	Up	--	192.168.90.3/24	<not set>	0090.21DB.1301	
FastEthernet0/1	Down	--	<not set>	<not set>	0090.21DB.1302	
Serial0/2/0	Up	--	10.0.0.9/30	<not set>	<not set>	
Serial0/2/1	Up	--	10.0.0.13/30	<not set>	<not set>	
Serial0/3/0	Down	--	<not set>	<not set>	<not set>	
Serial0/3/1	Down	--	<not set>	<not set>	<not set>	
Vlan1	Down	1	<not set>	<not set>	000A.4171.0B3E	
Physical Location: Intercity > Home City > Corporate Office > Main Wiring Closet > Rack > Router0						

Device Name: Router0(3)						
Custom Device Model: 2811 IOS15						
Hostname: Router						
Port	Link	VLAN	IP Address	IPv6 Address	MAC Address	
FastEthernet0/0	Up	--	192.168.10.3/24	<not set>	0007.EC89.6455	
FastEthernet0/0.2	Up	--	192.168.2.3/24	<not set>	0007.EC89.6455	
FastEthernet0/0.80	Up	--	192.168.80.3/24	<not set>	0007.EC89.6455	
FastEthernet0/1	Up	--	<not set>	<not set>	00D0.BA41.6821	
Serial0/2/0	Up	--	10.0.0.6/30	<not set>	<not set>	
Serial0/2/1	Up	--	10.0.0.14/30	<not set>	<not set>	
Serial0/3/0	Down	--	<not set>	<not set>	<not set>	
Serial0/3/1	Down	--	<not set>	<not set>	<not set>	
Vlan1	Up	1	<not set>	<not set>	00E0.8F8C.2BB6	
Physical Location: Intercity > Home City > Corporate Office > Main Wiring Closet > Rack > Router0(3)						


```
Router#show standby brief
```

```
      P indicates configured to preempt.
```

Interface	Grp	Pri	P	State	Active	Standby	Virtual IP
Fa0/0	1	110	P	Active	local	192.168.1.3	192.168.1.1
Fa	3	110	P	Active	local	192.168.3.3	192.168.3.1
Fa	4	110	P	Active	local	192.168.4.3	192.168.4.1
Fa	90	110	P	Active	local	192.168.90.3	192.168.90.1

```
Router#show standby brief
```

```
      P indicates configured to preempt.
```

Interface	Grp	Pri	P	State	Active	Standby	Virtual IP
Fa0/0	1	100	P	Standby	192.168.1.2	local	192.168.1.1
Fa	3	100	P	Standby	192.168.3.2	local	192.168.3.1
Fa	4	100	P	Standby	192.168.4.2	local	192.168.4.1
Fa	90	100	P	Standby	192.168.90.2	local	192.168.90.1

Configuration Interface Wan

```
#Router 1
```

```
conf t
interface se0/2/0
ip address 10.0.0.1 255.255.255.252
clock rate 6400
no shutdown
```

```
#Router 1bis
```

```
conf t
interface se0/2/0
ip address 10.0.0.2 255.255.255.252
no shutdown
```

```
#Configuration des routes en RIP
```

```
#Router 1
```

```
conf t
router rip
version 2
no auto-summary
network 192.168.1.0
network 192.168.90.0
network 192.168.3.0
network 192.168.4.0
network 10.0.0.0
```

```
#Router 1bis
```

```
conf t
router rip
version 2
no auto-summary
network 192.168.10.0
network 192.168.80.0
network 192.168.2.0
network 10.0.0.0
```

Network Address	
10.0.0.0	
192.168.1.0	
192.168.3.0	
192.168.4.0	
192.168.90.0	

Serial0/2/0

Port Status

☒ On

Duplex

☐ Full Duplex

Clock Rate

64000

IP Configuration

IPv4 Address

10.0.0.1

Subnet Mask

255.255.255.252

Tx Ring Limit

10

Les configuration Wan pour les deux autres routers Standby n'ont pas pu être abouti par manque de performance de la part de cisco packet tracer, donc schématiquement les routers seront en redondance Wan mais ne pourront pas vraiment l'être dans la pratique

Configurations DHCP/DNS/HTTP

DHCP

Interface	GigabitEthernet0	Service	☒ On	☐ Off
Pool Name	serverPool			
Default Gateway	192.168.80.1			
DNS Server	0.0.0.0			
Start IP Address :	192	168	80	7
Subnet Mask:	255	255	255	0
Maximum Number of Users :	249			
TFTP Server:	0.0.0.0			
WLC Address:	0.0.0.0			

Add		Save		Remove			
Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
RH	192.168....	192.168....	192.168....	255.255....	90	0.0.0.0	0.0.0.0
serverPool	192.168....	0.0.0.0	192.168....	255.255....	249	0.0.0.0	0.0.0.0

DHCP

Interface	GigabitEthernet0	Service	☒ On	☐ Off
Pool Name	serverPool			
Default Gateway	192.168.90.1			
DNS Server	192.168.90.4			
Start IP Address :	192	168	90	8
Subnet Mask:	255	255	255	0
Maximum Number of Users :	200			
TFTP Server:	0.0.0.0			
WLC Address:	0.0.0.0			

Add		Save		Remove			
Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
Finance	192.168....	192.168....	192.168....	255.255....	100	0.0.0.0	0.0.0.0
IT	192.168....	192.168....	192.168....	255.255....	100	0.0.0.0	0.0.0.0
serverPool	192.168....	192.168....	192.168....	255.255....	200	0.0.0.0	0.0.0.0

DHCP

Interface	GigabitEthernet0	Service	☒ On	☐ Off
Pool Name	serverPool			
Default Gateway	192.168.90.1			
DNS Server	192.168.90.4			
Start IP Address :	192	168	90	100
Subnet Mask:	255	255	255	0
Maximum Number of Users :	156			
TFTP Server:	0.0.0.0			
WLC Address:	0.0.0.0			

Add
Save
Remove

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
RH	192.168....	192.168....	192.168....	255.255....	156	0.0.0.0	0.0.0.0
Finance	192.168....	192.168....	192.168....	255.255....	156	0.0.0.0	0.0.0.0
serverPool	192.168....	192.168....	192.168....	255.255....	156	0.0.0.0	0.0.0.0
IT	192.168....	192.168....	192.168....	255.255....	156	0.0.0.0	0.0.0.0

Configuration SNMP/SYSLOG

Vu que Cisco Packet Tracer reste limité je ne vais pas pouvoir limité l'accès au SNMP et avoir un agent SNMP comme je l'aurai eu en vrai pareil pour le syslog qui aura des fonctionnalités réduites

#Ligne de commande pour la configuration SNMP

```
conf t
snmp-server community CHARLES RO
snmp-server community DINO RW
snmp-server host 192.168.80.5
snmp-server host 192.168.80.6
snmp-server enable traps
end
```

Ici DINO est le mot de passe pour les utilisateurs privés permettant d'avoir les droit de modification des appareils que le server SNMP gere et CHARLES est le mot de passe pour les utilisateurs publics n'ayant pas cette autorisation.

#Ligne de commande pour la configuration Syslog

```
conf t
logging host 192.168.80.5
logging host 192.168.80.6
logging trap informational
service timestamps log datetime msec localtime
```

exit

Configuration TFTP

#Cisco limite l'automatisation des envois de configuration aux serveurs TFTP.

enable

copy running-config tftp

#L'appareil demande l'IP du serveur :

192.168.90.6 192.168.90.7

#On nomme le dossier puis l'appareil enverra sa configuration au serveur

Relay et attributions IP des serveurs DHCP ____ Avant l'arrêt d'un des serveurs :

☒ DHCP	☐ Static	DHCP request successful.
IPv4 Address	192.168.3.6	
Subnet Mask	255.255.255.0	
Default Gateway	192.168.3.1	
DNS Server	192.168.90.4	

Figure 1: PCVLAN3

Après la reprise totale des attributions par l'autre serveur :

☒ DHCP	☐ Static	DHCP request successful.
IPv4 Address	192.168.3.6	
Subnet Mask	255.255.255.0	
Default Gateway	192.168.3.1	
DNS Server	192.168.90.4	

Test et configuration HTTP/DNS

DNS Service
☒ On
☐ Off

Resource Records

Name
Type
ARecord
▼

Address

Add
Save
Remove

No.	Name	Type	Detail
0	neoedurh	ARecord	192.168.90.5

Figure 2: ConfigDNS/RH

1.

DNS

DNS Service
☒ On
☐ Off

Resource Records

Name
Type
ARecord
▼

Address

Add
Save
Remove

No.	Name	Type	Detail
0	neoedu	ARecord	192.168.90.4

Figure 3: ConfigDNS

2.

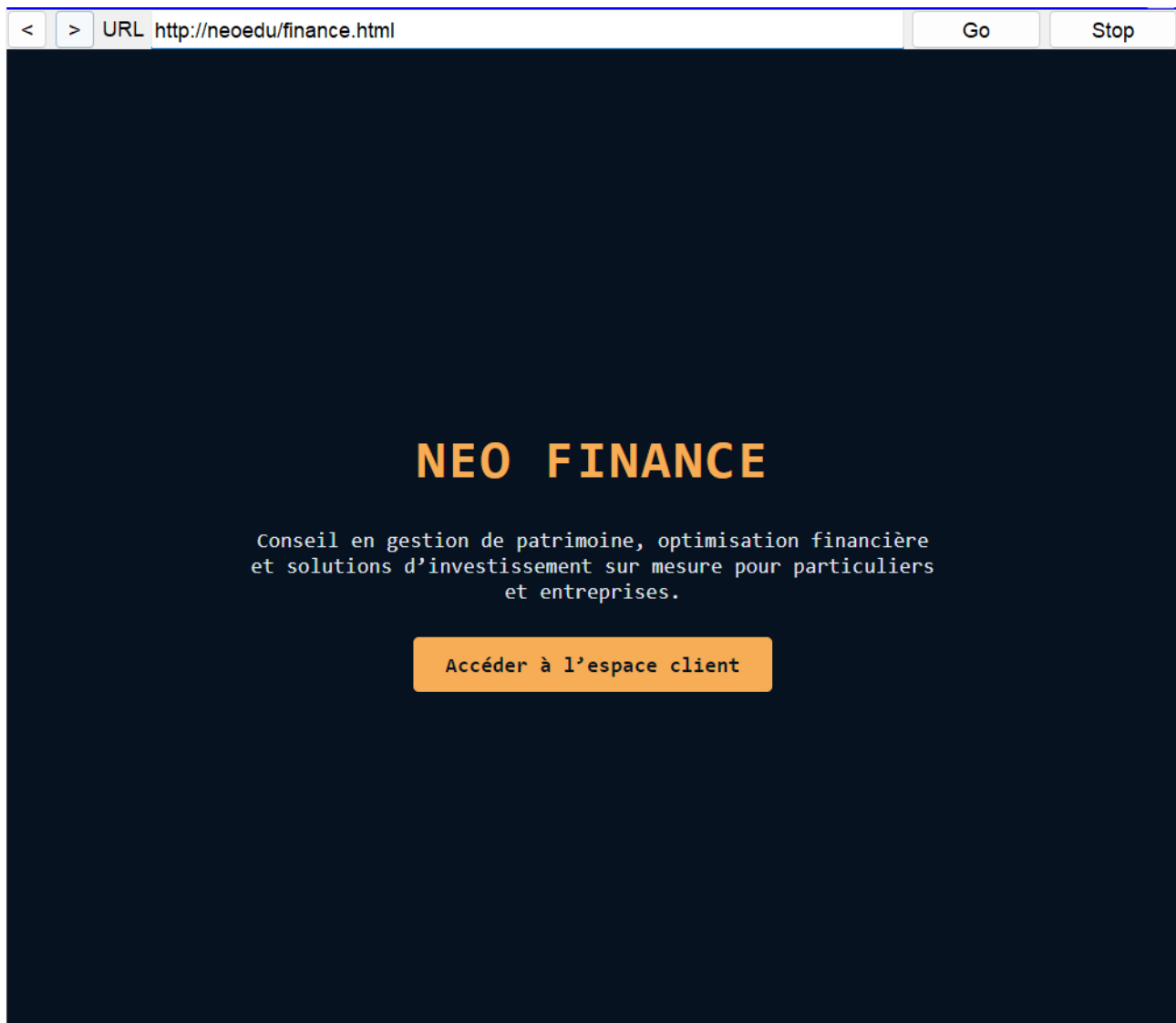


Figure 4: Page Finance

3.

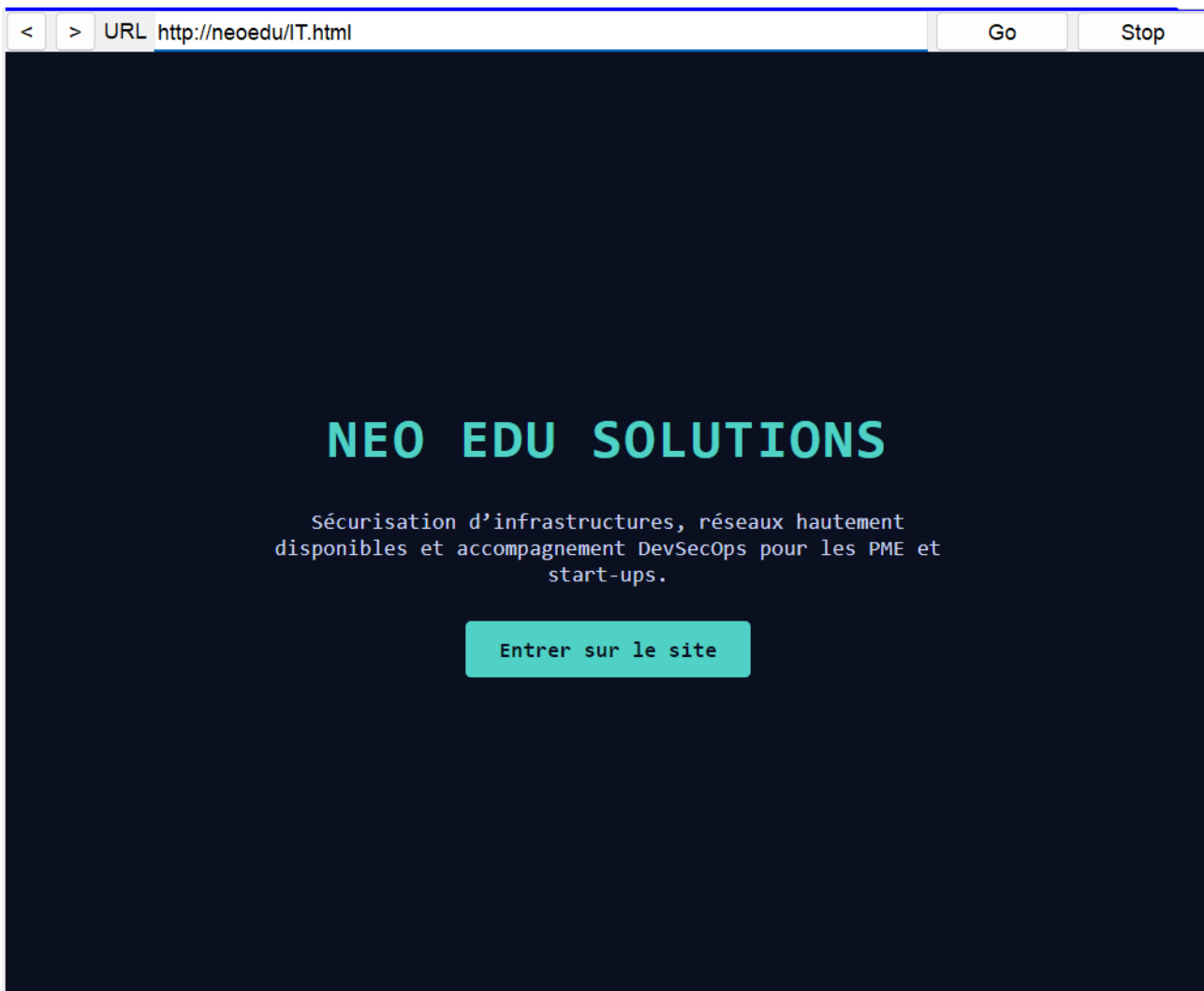
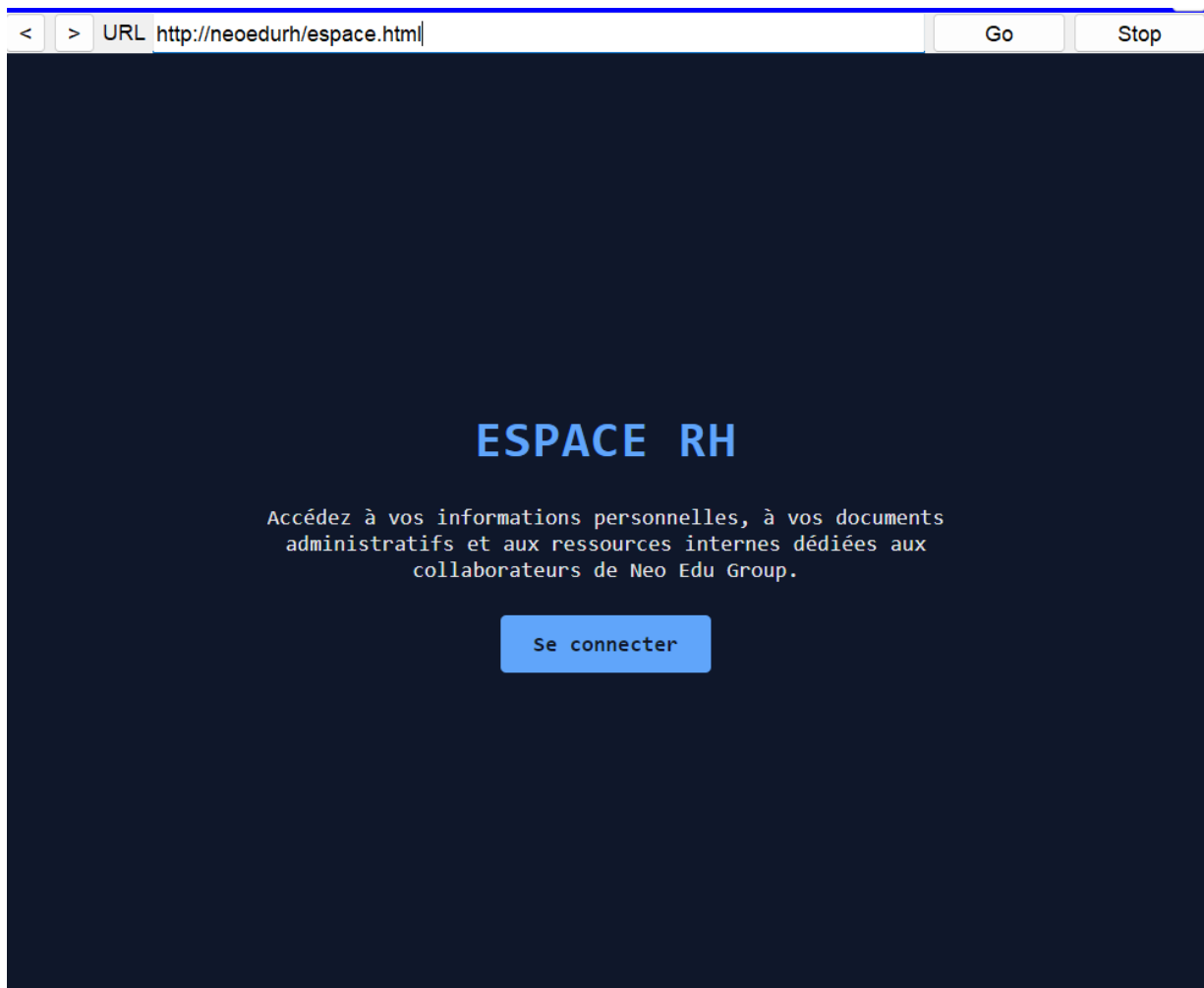


Figure 5: Page IT

4.



5.

Test TFTP

```
router1wan-running-config  
router2wan-running-config
```

Figure 6: Enregistrement de la configuration des router

Test SYSLOG et Standby ____

	Time	HostName	Message
1	03.01.1993 12:00:35.584 AM	192.168.80.3	%HSRP-6-...
2	03.01.1993 12:00:37.027 AM	10.0.0.13	%HSRP-6-...
3	03.01.1993 12:00:38.829 AM	192.168.80.3	%HSRP-6-...
4	03.01.1993 12:00:39.530 AM	192.168.80.3	%HSRP-6-...
5	03.01.1993 12:00:40.106 AM	10.0.0.13	%HSRP-6-...
6	03.01.1993 12:00:40.588 AM	10.0.0.13	%HSRP-6-...
7	03.01.1993 12:00:47.472 AM	192.168.80.3	%HSRP-6-...
8	03.01.1993 12:00:51.752 AM	10.0.0.13	%HSRP-6-...
9	03.01.1993 12:01:02.425 AM	10.0.0.13	%HSRP-6-...
10	03.01.1993 12:01:14.560 AM	10.0.0.13	%HSRP-6-...
11	03.01.1993 12:01:14.562 AM	10.0.0.13	%HSRP-6-...
12	03.01.1993 12:01:44.861 AM	10.0.0.13	%HSRP-6-...
13	03.01.1993 12:01:44.863 AM	10.0.0.13	%HSRP-6-...
14	03.01.1993 12:02:23.318 AM	10.0.0.13	%HSRP-6-...

Figure 7: Test des syslogs servers

9	03.01.1993 12:23:30.040 AM	192.168.80.3	Grp 2 state Standby -> Active
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Figure 8: Test des standby router

Test PING entre Vlan ____

Fire	Last Status	Source	Destination	Type	Color	Time(se	Periodic	Num	Edit	Delete
	Successful	Lapt...	RH2	ICMP		0.000	N	0	(edit)	(delete)
	Successful	SN...	Router0(2)	ICMP		0.000	N	1	(edit)	(delete)
	Successful	PCF...	PCIT	ICMP		0.000	N	2	(edit)	(delete)
	Successful	HTT...	PCFinance	ICMP		0.000	N	3	(edit)	(delete)
	Successful	PCIT	HTTP/DN...	ICMP		0.000	N	4	(edit)	(delete)
	Successful	DH...	PCFinance	ICMP		0.000	N	5	(edit)	(delete)
	Successful	DH...	PCIT	ICMP		0.000	N	6	(edit)	(delete)
	Successful	DH...	RH2	ICMP		0.000	N	7	(edit)	(delete)
	Successful	DH...	Router0(2)	ICMP		0.000	N	8	(edit)	(delete)
	Successful	HTT...	RH2	ICMP		0.000	N	9	(edit)	(delete)

Figure 9: TESTPING